

WARNING: THIS REVIEW IS NOT MEANT TO BE REFLECTIVE OF THE ACTUAL EXAM. THIS IS SIMPLY A SET OF SOME EXTRA PROBLEMS ON THE EXAM MATERIAL. IT IS NOT EXHAUSTIVE OF ALL TYPES OF MATERIAL OR QUESTIONS. YOU SHOULD USE THIS AS ONE STUDY TOOL, NOT YOUR ONLY STUDY TOOL!

1. One box contains 6 red balls and 4 green balls and a second box contains 7 red balls and 3 green balls. A ball is randomly chosen from the first box and placed unseen in the second box. Then a ball is randomly chosen from the second box and placed unseen in the first box. (A tree diagram might be helpful.)

a. What is the probability a red ball is drawn from box 2?

b. If a green ball is drawn from box 2, what is the probability a red ball was drawn from box 1?

2. Let V be the event that a computer contains a virus, and let W be the event that a computer contains a worm. Suppose $P(V) = 0.15$, $P(W) = 0.05$, and $P(V \cup W) = 0.17$. Round the answers to two decimal places.

a. Find the probability that the computer contains both a virus and a worm.

b. Find the probability that the computer contains neither a virus nor a worm.

c. Find the probability that the computer contains a virus but not a worm.

3. Political party preference in the United States depends in part on the age, income, and gender of the voter. A political scientist selects a large sample of registered voters. For each voter, she records gender, age in years, household income in dollars, and whether they voted for the Democratic, for the Republican, or for another candidate in the last congressional election.

What are all of the variables that will come from this dataset? Which are categorical and which are quantitative? Just list the variables in the correct column.

Categorical

Quantitative

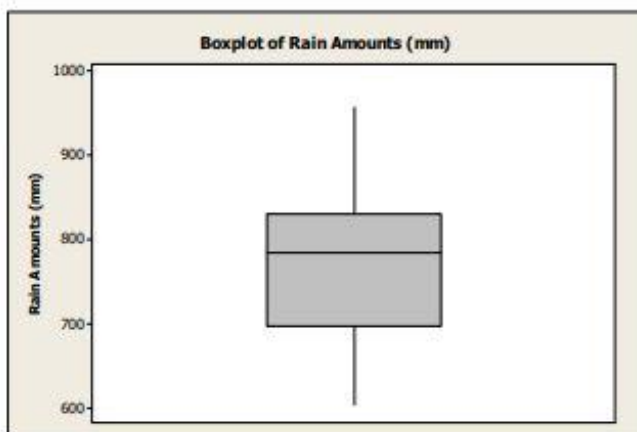
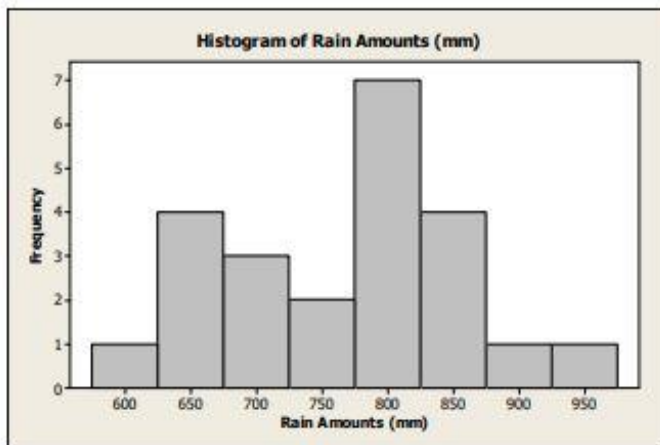
4. Hospitals typically require backup generators to provide electricity in the event of a power outage. Assume the emergency generators work independently of one another. The emergency backup generators fail 22% of the times when they are needed. A hospital has two backup generators. Find the probability that both generators fail during a power outage.

5. Here's some Minitab output on the monsoon rains (in mm) for 23 strong El Nino, the periodic warming of the Pacific Ocean west of South America, years between 1871 and 2004. Does it appear El Nino affects the monsoons? Let's examine.

Descriptive Statistics: Rain Amounts (mm)

Variable	Total Count	Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Rain Amounts (mm)	23	766.6	91.5	604.0	698.0	784.0	830.0	957.0

Variable	IQR
Rain Amounts (mm)	132.0



a. Discuss the five aspects of the overall distribution of this variable.

b. The average monsoon rainfall for all years from 1871 to 2004 is about 850 mm. What effect does El Nino appear to have on monsoon rains? Explain your answer.